PROJECT 06

Ball Mill Machine — Control Panel Wiring & Commissioning

Effective Engineering & Automation Pvt Ltd
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PROJECT SUMMARY

Wired and commissioned the complete electrical control panel for an industrial ball mill machine. The panel controls the 3-phase AC drive motor using a DOL starter with contactor, circuit protection, and operator push buttons. Everything was mounted on DIN rails, cables routed through ducts, and the panel passed full continuity, insulation, and function tests before handover.

OBJECTIVES

- Wire a complete 3-phase DOL motor starter panel to IEC standards.
- Include MCB (short circuit), RCCB (earth fault), and overload relay protection.
- Provide START, STOP, and EMERGENCY STOP operator controls.
- Route cables neatly through ducts with power and control cables separated.
- Pass full pre-commissioning test protocol before energising.

HOW IT WORKS

Incoming 3-phase supply → main isolator → RCCB → MCB → main contactor → overload relay → motor. Control circuit (230V AC): overload NC + e-stop NC + STOP NC + START NO → contactor coil + self-hold contact. Pressing START closes the contactor and motor runs; pressing STOP or e-stop opens the contactor. All wiring in ducts, cable ends ferrule-crimped and labelled.

KEY COMPONENTS

Component	Specification / Function
Main Isolator	[TO BE FILLED] A, 3-pole — panel isolation
RCCB	30 mA — earth fault protection
MCB	[TO BE FILLED] A, Type D — short circuit protection
3-Phase Contactor	[TO BE FILLED] A, 230V coil — motor switching
Overload Relay	Adjustable, set to motor FLC — thermal protection
START / STOP / E-Stop	Operator push buttons
DIN Rail + Ducts	Component mounting and cable management
Bootlace Ferrules	Crimped cable end terminations

TECHNICAL SPECIFICATIONS

Parameter	Value	Notes
Motor supply	3-phase AC, 400V, 50Hz	—
Motor FLC	[TO BE FILLED]	From motor nameplate
Control voltage	230V AC	Single phase from supply
Enclosure	Steel, IP54	—

MY CONTRIBUTION

- Selected all components from motor nameplate data and IEC standards.
- Planned panel layout to fit components with correct clearances.

- Wired complete 3-phase power circuit and 230V control circuit.
- Routed cables through ducts; crimped ferrules; applied cable labels.
- Ran full commissioning test (earth continuity, insulation, and function).

CHALLENGES & SOLUTIONS

Routing 3-phase power and control cables in a compact enclosure required careful planning to maintain minimum separation. An initial routing conflict was found and resolved by moving the control duct to the opposite side.

RESULTS

Panel fully wired, tested, and commissioned. All continuity, insulation (>1 MΩ), and function tests passed. Panel handed over with commissioning record. Good practical experience in 3-phase panel wiring and IEC-compliant commissioning.

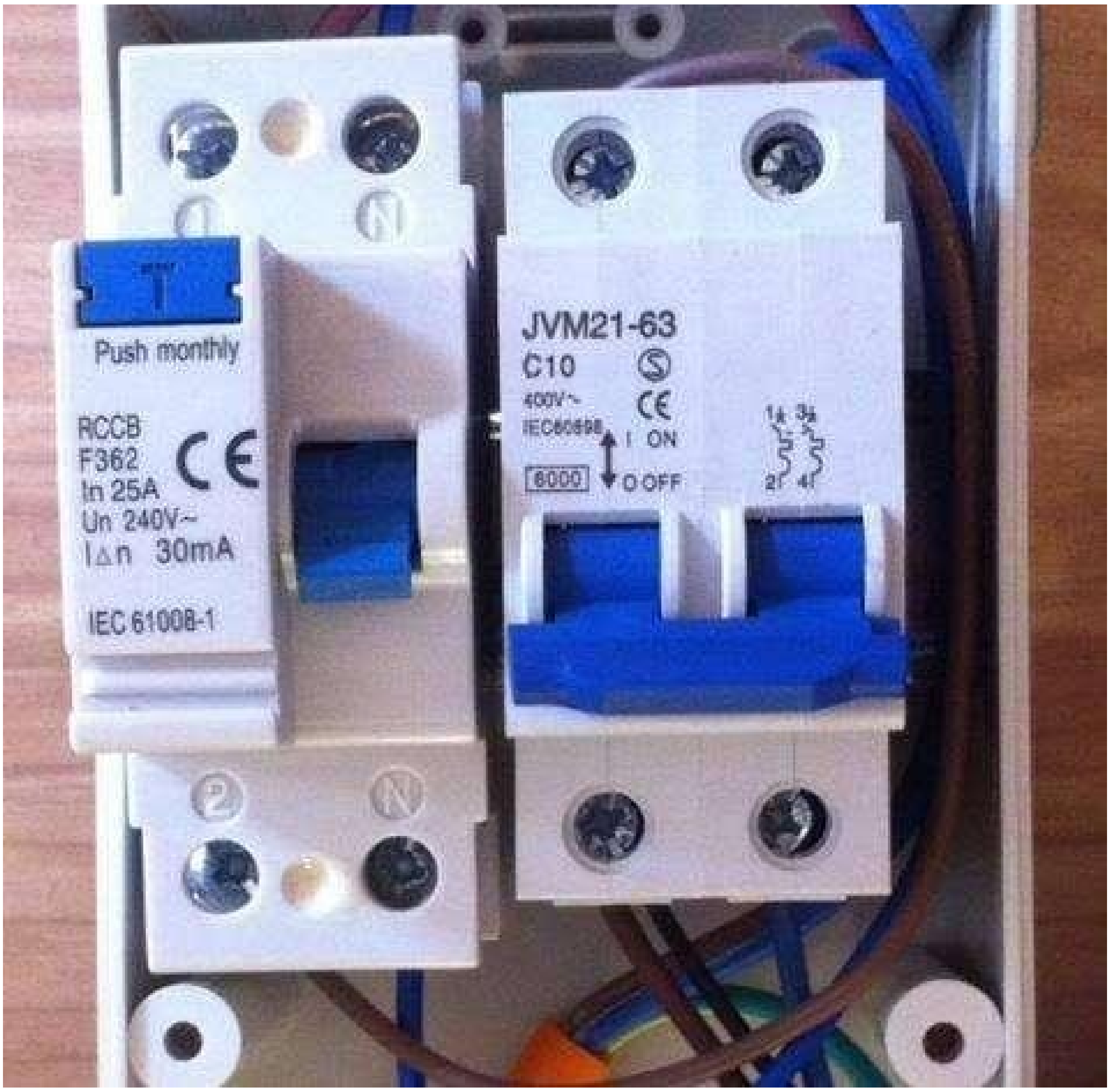
TOOLS & SOFTWARE USED

Tool / Software	Used For
Multimeter	Continuity, voltage, overload relay calibration
Insulation Tester (Megger)	500V insulation resistance test
Wire Crimping Tool	Ferrule crimping on all cable ends
Torque Screwdriver	Tightening terminals to spec

IMAGES & TECHNICAL DRAWINGS

Electrical connectors and terminal blocks

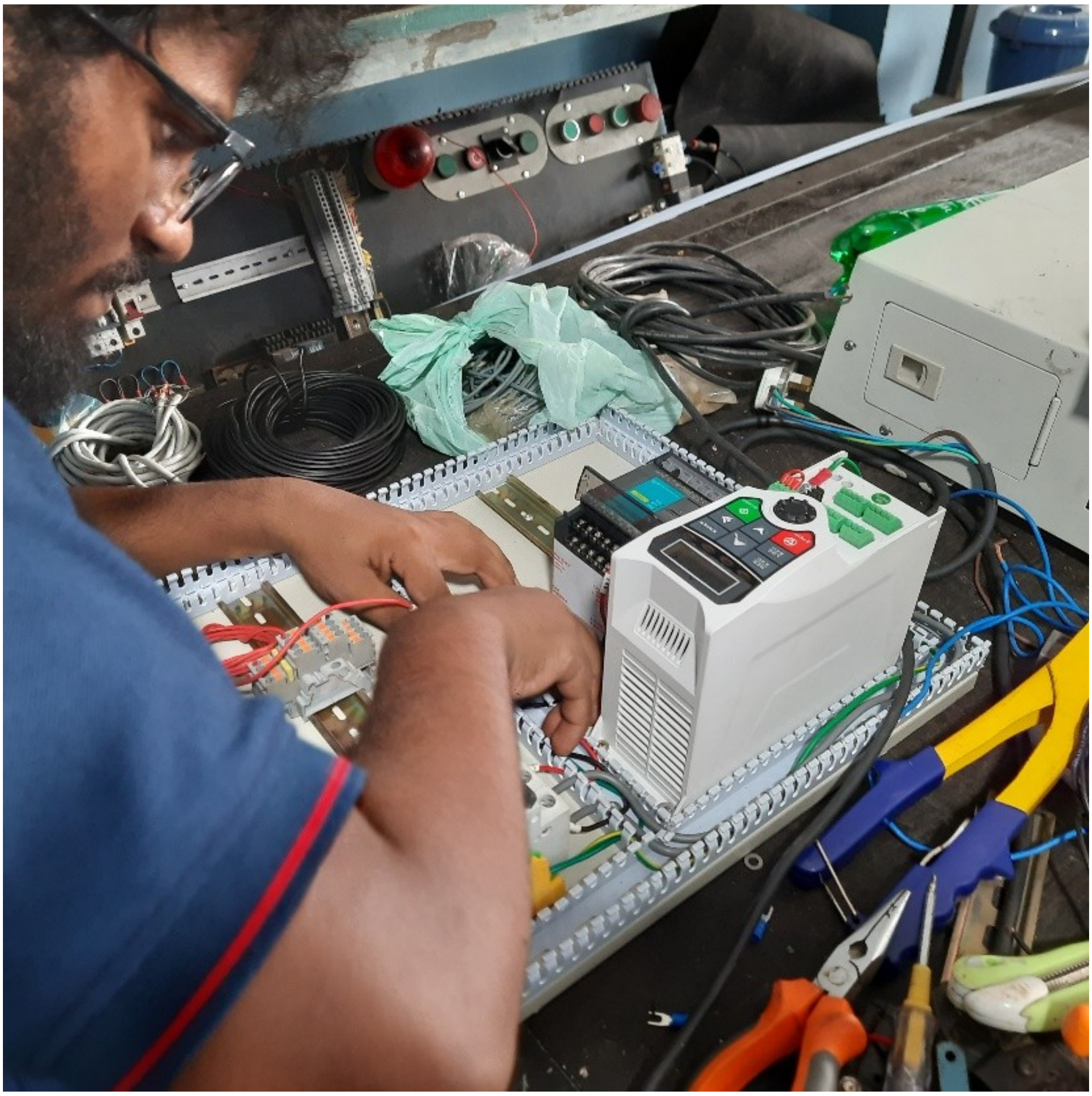
Relay modules



MCB and RCCB circuit protection devices

Lugs and cable connectors — crimped terminations

Ball mill panel wiring in progress



Ball mill panel wiring completed